Assignment 2 Java

1. Electricity Bill Calculator
Create a Java program to calculate a household electricity bill.

```
-->import java.util.Scanner;
class electricityBillDemo{
      public static void main(String args[]){
      Scanner sc = new Scanner(System.in);

     System.out.println("Enter Customer Name : ");
     String customerName = sc.nextLine();
     System.out.println("Enter Consumer Number : ");
     String consumerNumber = sc.nextLine();
     System.out.println("Enter Units : ");
     int units = sc.nextInt();

     double billAmt = calculateBill(units);

    System.out.println("Electricity Bill");
    System.out.println("Customer Name : " + customerName);
    System.out.println("Consumer Number : " + consumerNumber);
    System.out.println("Units : " + units);
    System.out.println("Total Electricity Bill : " + billAmt);

     }

     static double calculateBill(int units){
          double bill = 0;
          if(units <= 100){
               bill = units * 5;
          }else if (units <= 200){
               bill = 100 * 5 + (units - 100) * 7;
          }else {
               bill = 100 * 5 + 100 * 7 +(units -200)*10;
          }

          if(bill > 1500){
              bill = bill + (bill * 0.05);
          }
         return bill;
      }
}
```

---

2. Student Result Processing System
Create a program for a school to manage a student's marks.
-->

```
import java.util.Scanner;
class Assign{
   public static void main(String args[]){
    Scanner sc = new Scanner(System.in);

    System.out.println("Enter Student Name : ");
    String sname = sc.nextLine();
    System.out.println("Enter Student Roll Number : ");
    int rollNo = sc.nextInt();

    int totalMarks = 0;
    int subjects = 5;
    int marks[] = new int[5];

   for(int i =0; i< 5; i++){
    System.out.println("Enter Marks for subject :  "+ (i + 1) );
     marks[i] = sc.nextInt();
     totalMarks += marks[i];
   }

    double average =  totalMarks / 5;
    double percentage = (totalMarks / 500.0) * 100;

    char grade = calculateGrade(average);

    System.out.println("Student Results");
    System.out.println("Student Name : " + sname);
    System.out.println("Roll Number : " + rollNo);
    System.out.println("Total Marks : " + totalMarks);
    System.out.println("Average Marks : " + average);
    System.out.println("Percentage : " + percentage);
    System.out.println("Grade : " + grade);

  }

   static char calculateGrade(double average){
      if(average >= 90) return 'A';
      else if(average >= 75) return 'B';
      else if(average >= 60) return 'C';
      else if(average >= 40) return 'D';
      else return 'F';
```

```
  }
}
```

---

3. ATM Simulation
Create a simple ATM system.
Requirements
● Show a menu using switch-case:
1. Check Balance
2. Deposit Money
3. Withdraw Money
4. Exit
● Start with an initial balance.
● Prevent withdrawal if balance is insufficient.
● Continue showing the menu until the user chooses Exit.

```
--> import java.util.Scanner;
class ATMsystemDemo{

      public static void main(String args[]){
       Scanner sc = new Scanner(System.in);

       int balance = 2000 ;
       int choice;
       int amount;

       System.out.println("Enter Your Choice : 1. Check Balance 2. Deposite Money 3. Withdraw
Money 4.Exit");
       choice = sc.nextInt();
       switch(choice){
           case 1:
                if(balance >= 0){
                  System.out.println("Balance is : " + balance);
                 }else{
                    System.out.println("No Balance");
                 }
                 break;
           case 2:
                 System.out.println("Enter Amount to deposite: ");
                 amount = sc.nextInt();
                 balance += amount;
                 System.out.println("New balance is: "+balance);
                 break;
```

```
        case 3:
            System.out.println("Enter Amount to withdraw : ");
            amount = sc.nextInt();
            if (amount > balance){
             System.out.println("Insufficient Balance");
             }else{
              balance -= amount;
              System.out.println("New balance is: "+balance);
             }

           break;
        case 4 : sc.close();

        default :
           System.out.println("Invalid choice ");
      }
  }

}
```

---

4. Grocery Store Bill Generator
Create a billing system for a small grocery shop.
Requirements
● Accept item names, quantity, and price for multiple items.
● Store item totals in arrays.
● Calculate total bill amount.
● Apply discount:
○ 10% discount if bill is above Rs. 3000
● Print the final bill in proper format.

-->

```
import java.util.Scanner;
class Assign{
  public static void main(String args[]){
   Scanner sc = new Scanner(System.in);

  System.out.println("Enter number of items");
  int n = sc.nextInt();

   String[] itemNames = new String[n];
   int[] quantities = new int[n];
   double[] prices = new double[n];
```

```
    double totals[] = new double[n];

   double totalBill = 0.0;

   for(int i=0; i<n; i++){
        sc.nextLine();
     System.out.println("Enter item Name : ");
     itemNames[i]=sc.nextLine();
     System.out.println("Enter item quantity : ");
     quantities[i]=sc.nextInt();
     System.out.println("Enter item price : ");
     prices[i]=sc.nextDouble();

    totals[i]= quantities[i] * prices[i];
    totalBill += totals[i];
  }
   System.out.println("Bill");
  for(int i=0; i<n; i++){
       System.out.println(itemNames[i]+"\t"+ quantities[i]+"\t"+ prices[i]+"\t"+ totals[i]);
   }
   System.out.println("Total Bill : "+totalBill);
  if(totalBill>3000){
   totalBill *= 0.9;
   System.out.println("10% discount");
  }
 }
}
```

---

5. Employee Salary Slip Generator
Create a Java program to generate salary slips for employees.
Requirements
C-DAC Mumbai
● Accept employee name, ID, basic salary.
● Calculate:
○ HRA = 20% of basic salary
○ DA = 10% of basic salary
○ PF = 8% of basic salary
● Calculate net salary.
● Display a formatted salary slip.
-->

```
import java.util.Scanner;
class Assign{
   public static void main(String args[]){
```

```
        Scanner sc = new Scanner(System.in);

        System.out.println("Enter Employee Name : ");
        String name = sc.nextLine();
        System.out.println("Enter Employee Id : ");
        int Id = sc.nextInt();
        System.out.println("Enter Basic Salary : ");
        double basicSal = sc.nextDouble();

       double hra = basicSal * 0.2;
       double da = basicSal * 0.1;
       double pf = basicSal * 0.08;
       double grossSal = basicSal + hra + da;
       double netSal = grossSal - pf;

       System.out.println("HRA : "+ hra);
       System.out.println("DA : "+ da);
       System.out.println("PF : "+ pf);
       System.out.println("Net Salary : "+ netSal);
     }
}
```

---

6. Bus Ticket Booking System
Create a program for booking bus seats.
Requirements
● Assume a bus has 10 seats.
● Use an array to mark seats as booked or available.
● Show a menu:
1. View Available Seats
2. Book a Seat
3. Cancel a Seat
4. Exit
● Do not allow double booking.

```
-->import java.util.Scanner;
class BusTicketBookingDemo
{
   public static void main(String args[])
    {
          int seats[] = new int[10];
          Scanner sc = new Scanner(System.in);
          int choice = 0;
```

```
        System.out.println("Enter Your Choice : 1. View Available Seats 2. Book a Seat 3. Cancel
a Seat  4.Exit ");
        choice = sc.nextInt();
        switch(choice)
        {
          case 1:
                for(int i=0; i<10; i++)
                {
                     if(seats[i]==0)
                     {
                        System.out.println("Available Seat No: " +(i+1));
                     }else {
                           System.out.println("Seat No: " +(i+1)+" is booked");
                          }
                }
                break;

          case 2:
                System.out.println("Enter Seat Number (1 - 10 ): ");
                int a = sc.nextInt();
                if(a>=1 && a<=10 && seats[a-1]==0)
                  {
                    seats[a-1]=1;
                    System.out.println("seat Booked");
                  } else {
                    System.out.println("Already Booked");
                         }
                break;

          case 3:
               System.out.println("Enter Seat Number (1 - 10 ): ");
               int b = sc.nextInt();
               if(b>=1 && b<=10 && seats[b-1]==1)
               {
               seats[b-1]=0;
               System.out.println("seat cancelled");
               } else {
                System.out.println("Seat is available");
               }
               break;
          case 4: sc.close();
               break;
          default:
```

```
            System.out.println("Invalid choice");
        }
    }
}
```

---

7. Mobile Recharge System
Create a recharge application for prepaid mobile users.
Requirements
● Ask for mobile number and recharge amount.
● Show recharge plans using switch-case.
● Example:
○ Rs. 199 → 28 days validity
○ Rs. 399 → 56 days validity
○ Rs. 599 → 84 days validity
● Print recharge confirmation with plan details.
C-DAC Mumbai
-->

```
import java.util.Scanner;
class MobileRechargeSystem
{
        public static void main(String args[])
        {
            Scanner sc = new Scanner(System.in);

            System.out.println("Enter your mobile number : ");
            String mobNo = sc.nextLine();

            System.out.println("Recharge Plans are: ");
            System.out.println("1) Rs. 199 : 28 days validity ");
            System.out.println("2) Rs. 399 : 56 days validity ");
            System.out.println("3) Rs. 599 : 84 days validity ");

            System.out.println("Enter Recharge Amount");
            int Amt = sc.nextInt();

            /*System.out.println("Enter your Choice: 1. 2. 3." ); ****
              int choice = sc.nextInt();*/

             switch(Amt)
             {
                 case 199 : if(Amt == 199)
```

```
                        System.out.println(" Recharge Successful !\n Plan Name : 199\n Mobile
Number : "+mobNo);
                    break;
                case 399 : if(Amt == 399)
                        System.out.println(" Recharge Successful !\n Plan Name : 399\n Mobile
Number : "+mobNo);
                    break;
                case 599 : if(Amt == 599)
                        System.out.println(" Recharge Successful !\n Plan Name : 599\n Mobile
Number : "+mobNo);
                    break;
                default : System.out.println("Invalid Plan");
            }
        }
}
```

---

8. Hospital Patient Record System
Create a simple patient management program.
Requirements
● Create a class Patient with:
○ patient ID
○ patient name
○ age
○ disease
● Create objects for at least 3 patients.
● Display patient details.
● Add a method to check whether the patient is a senior citizen (age 60 or above).
-->

```
class PatientRecordDemo
{
  int PatientId;
  String PatientName;
  int age;
  String disease;

  PatientRecordDemo()
  {

  }
  PatientRecordDemo(int PatientId, String PatientName, int age, String disease)
  {
      this.PatientId = PatientId;
```

```
        this.PatientName = PatientName;
        this.age = age;
        this.disease = disease;
    }

   void print()
   {
     System.out.println("Patient Id: "+PatientId+" Patient Name : "+PatientName+" Age : "+age+"
Disease : "+disease);
     if(age >= 60)
         System.out.println(PatientName+" is a Senior citizen");
   }
}
class ObjectDemo
{
      public static void main(String args[])
     {
      PatientRecordDemo p1 = new PatientRecordDemo(1,"Raha", 50,"Fever");
      p1.print();
      PatientRecordDemo p2 = new PatientRecordDemo(2,"John", 60,"Dengue");
      p2.print();
      PatientRecordDemo p3 = new PatientRecordDemo(3,"Sana", 20,"Maleria");
      p3.print();

     }
}
```

---

9. Daily Expense Tracker
Create a program to track a person’s daily expenses.
Requirements
● Accept expenses for 7 days in an array.
● Calculate:
○ total weekly expense
○ average expense
○ highest expense
○ lowest expense
● Display the day on which the highest expense occurred.
-->

```
 import java.util.Scanner;
class ExpensesDemo
{
    public static void main(String args[])
    {
```

```
        Scanner sc = new Scanner(System.in);

        System.out.println("Enter expenses of 7 days");
        int arr[] = new int[7];
        String days[] =
{"Sunday","Monday","Tuesday","Wednesday","Thursday","Friday","Saturday"};
        for(int i =0; i<arr.length; i++)
        {
          arr[i] = sc.nextInt();
        }
        System.out.print("Expenses are: ");
        for(int i =0; i<arr.length; i++)
        {
          System.out.println(arr[i]+"\t");
        }
        int Total =0;
        for (int i : arr)
        {
           Total += i;
        }
        System.out.println("total: "+Total);

        int Average = Total / arr.length;
        System.out.println("Average : "+Average);

        int max = arr[0];
        int min = arr[0];
        int maxDay = 0;
        int minDay = 0;

        for (int i = 0; i < arr.length; i++)
         {
          Total += arr[i];
          if (arr[i] > max) {
              max = arr[i];
              maxDay = i;
          }
          if (arr[i] < min) {
              min = arr[i];
              minDay = i;
          }
         }
         System.out.println("Highest Expense: "+max);
         System.out.println("Day of Highest Expense: "+days[maxDay]);
```

```
        System.out.println("Lowest Expense: "+min);
        System.out.println("Day of Lowest Expense: "+days[minDay]);
    }
}
```

---

10. Restaurant Order Management System
Create a program for restaurant management.
Requirements
● Display a food menu with price list.
● Let the user order multiple items.
● Use loop and switch-case for repeated ordering.
C-DAC Mumbai
● Calculate total amount.
● Add GST of 5%.
● Print final bill.
-->

```
import java.util.Scanner;
class RestaurantManagementDemo
{

    public static void main(String args[]){

        Scanner sc = new Scanner(System.in);

      System.out.println(" Menu");
      System.out.println("1) Maggie     - Rs. 50");
      System.out.println("2) Vada Pav   - Rs. 20");
      System.out.println("3. Pasta      - Rs. 180");
      System.out.println("4. Bhel       - Rs. 80");
      System.out.println("5. Lassi      - Rs. 40");
          int choice = 0;
          int quantity = 0;
          String repeat = " ";
      double totalAmount = 0;
      do {
         System.out.println("Enter your choice (1-5): ");
         choice = sc.nextInt();
         System.out.println("Enter quantity: ");
         quantity = sc.nextInt();

         switch (choice) {
            case 1:
```

```
                totalAmount += 50 * quantity;
                break;
            case 2:
                totalAmount += 20 * quantity;
                break;
            case 3:
                totalAmount += 180 * quantity;
                break;
            case 4:
                totalAmount += 80 * quantity;
                break;
            case 5:
                totalAmount += 40 * quantity;
                break;
            default:
                System.out.println("Invalid choice!");
        }


        System.out.println("Do you want to order more items? (yes/no): ");
        repeat = sc.next();
          }while(repeat.equalsIgnoreCase("yes"));

      double gst = totalAmount * 0.05;
      double finalBill = totalAmount + gst;

      System.out.println(" Final Bill");
      System.out.println("Total Amount: Rs. " + totalAmount);
      System.out.println("GST (5%): Rs. " + gst);
      System.out.println("Final Bill: Rs. " + finalBill);

    }
}
```

---

11. Calculator System (Overloaded Methods)
Create a Java program for a calculator:
Requirements:
● Create methods named calculate
● Overload them for:
○ int + int
○ double + double
○ int + int + int

● Call all methods from main
● Display results clearly
-->

```
class calculator
{
   void calculate(int a, int b)
   {
      res1 = a + b;
      System.out.println("Result of int + int : "+res1)
   }
   void calculate(double a, double b)
   {
      res2 = a + b;
      System.out.println("Result of double + double : "+res2)

   }
   void calculate(int a, int b, int c)
   {
      res3 = a + b + c;
      System.out.println("Result of int + int + int : "+res1)

   }

}


class CalculatorDemo
{
       public static void main(String samruddhi[])
      {
         calculator cl = new calculator();
         c1.calculate(10, 20);
         c1.calculate(10.0, 20.0);
         c1.calculate(10, 20, 30);
      }
}
```

---

12. Student Counter System
Create a class Student:
Requirements:
● Instance variables: name, rollNo
● Static variable: totalStudents
● Increment totalStudents every time object is created

● Display total number of students
Bonus: Add a static method to display count

-->

```
class Student
{
   int rollNo;
   String name;
   static int totalStudent = 0;

   Student()
   {

   }
   Student(int rollNo, String name)
   {
      this.rollNo = rollNo;
      this.name =name;
      totalStudent++;
   }
   void display()
   {

      System.out.println("Roll Number: "+rollNo+" Name: "+name);
   }
   static void show()
   {
      System.out.println("Total Students: "+totalStudent);
   }
}


class StudentCounterDemo
{
        public static void main(String samruddhi[]){
   Student st1 = new Student(1,"Gauri");
   st1.display();
   Student st2 = new Student(2,"Purva");
   st2.display();
   Student.show();
   }
}
```

---

13. Bank Account System
Create a class BankAccount:
Requirements:
● Variables: accountNumber, name, balance
● Use:
C-DAC Mumbai
○ Default constructor
○ Parameterized constructor
● Display account details
● Create multiple objects using different constructors
-->

```
class BankAccount
{
  int accountNumber;
  String name;
  double balance;

  BankAccount()
  {

  }
  BankAccount(int accountNumber, String name, double balance)
  {
     this.accountNumber = accountNumber;
     this.name = name;
     this.balance = balance;
  }

  void display()
  {
    System.out.println("Account Details ");
    System.out.println("Account Number : "+accountNumber);
    System.out.println("Name : "+ name);
    System.out.println("Bank Balance : "+balance);
  }


}

class BankDemo
{
    public static void main(String args[])
    {
       BankAccount b1 = new BankAccount(12321,"Sandhya",20000);
```

```
        BankAccount b2 = new BankAccount();
        BankAccount b3 = new BankAccount(12355,"Sneha",50000);
        BankAccount b4 = new BankAccount(54465,"Nikhil",20000);
        b1.display();
        b2.display();
        b3.display();
        b4.display();
    }
}
```

---

14. Salary Conversion System
Create a program:
Requirements:
● Take the salary as double
● Convert it into:
○ int (explicit casting)
○ float (implicit casting)
● Show the precision loss clearly
-->

```
class salaryConversionDemo
{
    public static void main (String args[])
    {
        double salary = 586486.6346;
        System.out.println("Salary in double: "+salary);

        int s2 = (int)salary;
        System.out.println("Salary in int: "+s2);
        System.out.println("Precision loss due to Double into int: "+(salary - s2));

        float s3 = (float)salary;
        System.out.println("Salary in float: "+s3);
        System.out.println("Precision loss due to Float into int: "+(salary - s3));
    }
}
```

---

15. Student Marks Table
Create a program:
Requirements:
● Store marks of 3 students in 3 subjects using 2D array
● Calculate:
○ Total per student

○ Average marks
● Display in table format
-->

```
 import java.util.Scanner;
class Student
{
    public static void main(String args[])
    {
        int arr[][] = new int[3][3];
        Scanner sc = new Scanner(System.in);
        for(int i=0; i<3; i++)
        {
            System.out.println("Enter marks of students: ");
            for(int j=0; j<3; j++)
            {
               arr[i][j]= sc.nextInt();
            }
        }
        for(int i=0; i<3; i++)
        {   int total =0;
            for(int j=0; j<3; j++)
            {
               total += arr[i][j];
            }
            double avg = total/3.0;
            System.out.println("Student : "+(i+1)+" | Total : "+total+" | Average : "+avg);
        }


    }

}
```

---

16. Weekly Work Hours
Create a program:
Requirements:
● Each employee works different number of days
● Use jagged array to store hours
● Display all employee work hours
C-DAC Mumbai
-->

```
class EmployeehoursDemo
{
```

```
    public static void main(String args[])
   {
          int arr[][] ={{1, 2},
                    {5,6,8},
                    {5,3},
                    {5,6,3,8}};

         /* System.out.println("Employee 1 : "+(arr[0][0]+arr[0][1]));
          System.out.println("Employee 2 : "+(arr[1][0]+arr[1][1]+arr[1][2]));
          System.out.println("Employee 3 : "+(arr[2][0]+arr[2][1]));
          System.out.println("Employee 4 : "+(arr[3][0]+arr[3][1]));*/

         for (int i = 0; i < arr.length; i++) {           //length of column
          int totalHours = 0;
          for (int j = 0; j < arr[i].length; j++) {        // length of row
              totalHours += arr[i][j];
          }
          System.out.println("Employee No " + (i+1) + " : " + totalHours);
      }
    }
}
```

---

17. Scope Demonstration Program
Create a program:
Requirements:
● Declare variable inside:
○ Method
○ Loop
○ Block
● Try accessing outside scope (show error via comments)
● Print valid usages
-->

```
class Student
{
      {
        int a = 3;
        System.out.println("Inside Block : "+a);
      }
      // System.out.println(a); // error: <identifier> expected

      void display()
      {
        int b = 5;
```

```
        System.out.println("Inside method : "+b);
       }
       // System.out.println(b);  // error: <identifier> expected

       void loopDemo()
       {
           for(int c=0; c<1; c++)
           {
              System.out.println("Inside the Loop : "+c);
           }
           //System.out.println(c);  // error: cannot find symbol
       }

}
class Assign
{
       public static void main(String args[])
      {

         Student s = new Student();
         s.display();
         s.loopDemo();

      }
}
```

---

18. Expression Evaluator
Create a program:
Requirements:
● Take 3 numbers
● Perform:
○ (a + b) * c
○ a > b && b > c
○ a % b
● Show results with explanation
-->

```
 import java.util.Scanner;
class ExpressionDemo
{
    public static void main(String args[])
    {
       Scanner sc = new Scanner(System.in);
```

```
      System.out.println("Enter 3 Numbers ");
        int a = sc.nextInt();
        int b = sc.nextInt();
        int c = sc.nextInt();


         int d = (a + b) * c;
         System.out.println("First expression in parenthisis evaluate then multiplication is evaluate
and gives Result ");
         System.out.println("Result = "+d);

         Boolean e = a > b && b > c;
         System.out.println("If both the Conditions True &&(logical AND) return true else return
False  ");
         System.out.println("Result : "+e);

         int f = a % b;
         System.out.println("%(modulus) returns Remainder of a divided by b ");
         System.out.println("Result : "+f);

   }

}
```

---

19. Employee ID Generator
Create a class Employee:
Requirements:
● Static variable: counter
● Each employee gets auto-incremented ID
● Use constructor to assign ID
● Display employee details
C-DAC Mumbai
-->

```
class IdDemo
{
   static int counter = 1 ; // If 0 then Employee Id starts with 0;
   int Id;
   String name;

   IdDemo()
   {

   }
```

```
    IdDemo(String name)
    {
        this.Id = counter++;
        this.name = name;
    }

    void display()
    {
          System.out.println("Employee Id: "+Id+" |  Name : "+name);
    }

}

class EmployeeDemo
{
    public static void main(String args[])
    {
         IdDemo e1 = new IdDemo("Mira");
         IdDemo e2 = new IdDemo("Santosh");

         e1.display();
         e2.display();
   }
}
```

---

20. Library System
Create two classes:
Class 1: Book
● bookId, bookName
Class 2: Library
● Method to display books
Requirements:
● Create multiple books
● Pass objects to library
● Display all books
C-DAC
-->

```
import java.util.ArrayList;
class Book
{
   int bookId ;
   String name ;
```

```
  Book()
  {

  }

  Book(int bookId, String name)
  {
      this.bookId = bookId;
      this.name = name;
  }
  void display()
  {
      System.out.println("Book Id : "+bookId+" , Book Name : "+name);
  }

}

class Library
{
       ArrayList<Book>books;

      Library()
      {
           books = new ArrayList<>();
      }
       void addBook(Book b)
      {
       books.add(b);
      }

      void displayAllBooks()
      {
          System.out.println("Books in Library:");
          for (Book b : books)
         {
            b.display();
         }
      }
}

class LibraryBooksDemo
{
    public static void main(String args[])
    {
```

```
        Book b1 = new Book(101, "C++ Concepts");
        Book b2 = new Book(102, "Data Structure");
        Book b3 = new Book(103, " Database Technologies");

        Library l1 = new Library();

        l1.addBook(b1);
        l1.addBook(b2);
        l1.addBook(b3);

        l1.displayAllBooks();
    }

}
```